

DUMMY VARIABLE

How to deal with qualitative, and not quantitative, predictors?

1. The situation with two levels:

The simplest case is the situation with two levels. A binary variable C with two levels (for example, a and b) can be encoded as

$$D = \begin{cases} 1 & \text{if } C = b \\ 0 & \text{if } C = a \end{cases}$$

D now is a dummy variable: it codifies with zeros and ones the two possible levels of the categorical variable.

Example : C could be gender, which has levels male and female. The dummy variable associated is $D = 0$ if the gender is male and $D = 1$ if the gender is female.

The advantage of this dummification is its interpretability in regression models. Since level a corresponds to 0, it can be seen as the reference level to which level b is compared. This is the key point in dummification: set one level as the reference and codify the rest as departures from it with ones. The previous interpretation translates easily to the linear model. Assume that the dummy variable D is available together with other predictors $X_1, \dots, X_{p-1}$. Then:

$$\mathbb{E}[Y|X_1 = x_1, \dots, X_{p-1} = x_{p-1}, D = d] = \beta_0 + \beta_1 X_1 + \dots + \beta_{p-1} X_{p-1} + \beta_p D.$$

The coefficient associated to D is easily interpretable: β_p is the increment in mean of Y associated to changing $D = 0$ (reference) to $D = 1$, while the rest of the predictors are fixed. Or in other words, β_p is the increment in mean of Y associated to changing the level of the categorical variable from a to b.

R does the dummification automatically (translates a categorical variable C into its dummy version D) if it detects that a factor variable is present in the regression model.

2. The situation with 3 levels:

For example, a categorical variable C with levels a, b, and c. If we take a as the reference

level, this variable can be represented by two dummy variables:

$$D_1 = \begin{cases} 1 & \text{if } C = b \\ 0 & \text{if } C \neq b \end{cases}$$

$$D_2 = \begin{cases} 1 & \text{if } C = c \\ 0 & \text{if } C \neq c \end{cases}$$

Then

- $C = a$ is represented by $(D_1, D_2) = (0, 0)$
- $C = b$ is represented by $(D_1, D_2) = (1, 0)$
- $C = c$ is represented by $(D_1, D_2) = (0, 1)$

The interpretation of the regression models with the presence of D_1 and D_2 is very similar to the one before. For example, for the linear model, the coefficient associated to D_1 gives the increment in mean of Y when the category of C changes from a to b , and the coefficient for D_2 gives the increment in mean of Y when C changes from a to c .

3. The case with more than two levels:

In general, if we have a categorical variable C with J levels, then the previous process is iterated and the number of dummy variables required to encode C is $J - 1$. Again, R does the dummification automatically if it detects that a factor variable is present in the regression model.